THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 60°	Complementary angles add up to 90°.
2	**(a)** even	Use the even/odd rules.
3	**(b)** Yes	Two sides together must be longer than the third.
4	**(d)** $\frac{2}{5}$	Multiply numerators and multiply denominators, then simplify.
5	**(b)** 130°	Supplementary angles add up to 180°.
6	**(c)** 11	Add all the digits.
7	**(a)** 70°	The three angles add up to 180°.
8	**(b)** 8	'Of' means multiply.
9	**(b)** 40°	Vertically opposite angles are equal.
10	**(c)** 5743	Write the digits in the opposite order.
11	**(d)** equilateral	Recall the definition.
12	**(d)** $\frac{7}{3}$	Turn the fraction upside down.
13	**(d)** 120°	A linear pair adds up to 180°.
14	**(c)** 1331	A palindrome reads the same from both ends.
15	**(a)** 95°	Exterior angle = sum of the two interior opposite angles.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Complement $= 90^\circ - 35^\circ = 55^\circ$ (1); Supplement $= 180^\circ - 110^\circ = 70^\circ$ (1).	1+1
17	Apply the rule to each number in turn; Sequence: $6, 3, 10, 5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 8 steps (1).	1+1
18	Two shorter sides: $6 + 8 = 14$; Compare with the longest side 10 (1); Yes, 14 > 10 (1).	1+1
19	$\frac{3}{5} \times \frac{10}{21} = \frac{3\times10}{5\times21}$; $= \frac{30}{105}$ (1); Lowest terms: $\frac{2}{7}$ (1).	1+1
20	Vertically opposite angle $= 65^\circ$ (1); The other two angles form a linear pair with it: $180^\circ - 65^\circ = 115^\circ$ each (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Sum $= 180^\circ$ and the number of parts $= 3$ (1); One part $= 60^\circ$ (1); The angles are 60° and 120° (1).	1+1+1
22	Largest $= 5432$ (1); Smallest $= 2345$ (digits arranged in increasing order) (1); Difference $= 5432 - 2345 = 3087$ (1).	1+1+1
23	Total parts $= 6$; one part $= 180^\circ \div 6 = 30^\circ$ (1); Angles 30°, 60°, 90° (1); Check: 180° (1).	1+1+1
24	Improper form: $2\frac{1}{3} = \frac{7}{3}$ (1); $\frac{7}{3} \times \frac{3}{5} = \frac{21}{15}$ (1); Answer $1\frac{2}{5}$ (1).	1+1+1
	OR: $x = 180 - 110 = 70$ (1); The angle 70° is acute (2).	1+2
25	Middle number $= 48 \div 3 = 16$ (1); The numbers are 15, 16 and 17 (1); Check: $15 + 16 + 17 = 48$ (1).	1+1+1
	OR: Base angles are equal; their sum $= 180^\circ - 30^\circ = 150^\circ$ (1); Each base angle $= 75^\circ$ (1); Check: the three angles add up to 180° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Vertically opposite angle $= 35^\circ$ (1); The other two angles are 145° each (linear pair) (1); Sum $= 35 + 35 + 145 + 145 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Reverse $= 62$ (1); Sum $= 26 + 62 = 88$ (1); $88 = 11 \times 8$ (1); Divisible by 11; quotient $= 8$ (2).	1+1+1+2
27	$5 + 5 = 10 > 8$, so the triangle exists (1); It is isosceles (1); Perimeter $= 5 + 5 + 8$ (1); $= 18$ cm (1); Final statement (1).	1+1+1+1+1
	OR: At first $\frac{3}{4} \times 120 = 90$ L (1); Used $\frac{2}{3} \times 90 = 60$ L (1); Left $= 90 - 60$ (1); $= 30$ L (1); State all three answers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{4}$	Dividing by n means multiplying by $\frac{1}{n}$.
2	**(d)** parallel lines	Recall the definition.
3	**(a)** 1827	A number is divisible by 9 if its digit sum is.
4	**(a)** 70°	Base angles are equal and the three angles add up to 180°.
5	**(a)** smaller than each of the fractions	Recall the idea from the chapter.
6	**(b)** 55°	Corresponding angles are equal when the lines are parallel.
7	**(d)** 22	Check whether the number is even or odd.
8	**(b)** 21	Add the three sides.
9	**(b)** $\frac{7}{3}$	Whole $\times$ denominator + numerator, over the denominator.
10	**(d)** 48°	Alternate interior angles are equal when the lines are parallel.
11	**(b)** 36	The sum of the first n odd numbers is n^2.
12	**(c)** SSS	Compare the matching parts.
13	**(a)** $\frac{1}{3}$	Area = length $\times$ breadth.
14	**(d)** 118°	Interior angles on the same side add up to 180°.
15	**(d)** 74	Value $= 10\times$ tens $+$ units.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Add the number to its reverse; $29 + 92 = 121$ (1); Palindrome obtained: 121 (1).	1+1
17	Third $= 180^\circ - (48^\circ + 72^\circ); = 60^\circ$ (1); The triangle is acute-angled (1).	1+1
18	Reciprocal of $\frac{9}{16}$ is $\frac{16}{9}$; $\frac{3}{4} \times \frac{16}{9} = \frac{48}{36}$ (1); Answer $1\frac{1}{3}$ (1).	1+1
19	Vertically opposite angle $= 70^\circ$ (1); Linear pair: $180^\circ - 70^\circ = 110^\circ$ (1).	1+1
20	Digit sum $= 18$; Divisible by 3: yes (digit sum is a multiple of 3) (1); Divisible by 9: yes (digit sum is a multiple of 9) (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Pieces $= n \div \frac{3}{4}$ (1); $6 \times \frac{4}{3} = \frac{24}{3}$ (1); Pieces $= 8$ (1).	1+1+1
22	(i) Alternate interior angles are equal: 52° (1); (ii) Interior angles on the same side add up to 180°: 128° (2).	1+2
23	13×7 is odd (odd $\times$ odd) (1); 4 is even (1); The sum is odd (1).	1+1+1
24	Exterior $=$ sum of interior opposite angles (1); Other $= 120^\circ - 45^\circ = 75^\circ$ (1); Adjacent interior angle $= 180^\circ - 120^\circ = 60^\circ$ (1).	1+1+1
	OR: Area $= \frac{3}{4} \times \frac{2}{3}$; Area $= \frac{1}{2}$ m^2 (1); Perimeter $= 2(\frac{3}{4} + \frac{2}{3})$; $= 2 \times \frac{17}{12}$ (1); Perimeter $= 2\frac{5}{6}$ m (1).	1+1+1
25	Sum $= 180^\circ$ and the number of parts $= 6$ (1); One part $= 30^\circ$ (1); The angles are 30° and 150° (1).	1+1+1
	OR: Largest $= 8531$ (1); Smallest $= 1358$ (digits arranged in increasing order) (1); Difference $= 8531 - 1358 = 7173$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Let the complement be y; the angle is $2y$ and $2y + y = 90$ (1); $y = 30$ (2); The angle is 60° and its complement is 30° (2).	1+2+2
	OR: Sequence: $9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Steps $= 19$ (1); Largest number $= 52$ (1); Even numbers: 13 (1); Ends in the loop 4, 2, 1 (1).	1+1+1+1+1

27	$\angle C = 180^\circ - 40^\circ - 65^\circ = 75^\circ$ (1); Classification: acute-angled (1); Exterior angle at C $= 180^\circ - 75^\circ$ (1); $= 105^\circ$ (1); It equals $\angle A + \angle B = 40^\circ + 65^\circ$ (1).	1+1+1+1+1
	OR: Cakes $= 5 \div \frac{3}{4} = 6\frac{2}{3}$ (1); Full cakes $= 6$ (1); Flour used $= 6 \times \frac{3}{4} = 4\frac{1}{2}$ kg (1); Left over $= \frac{1}{2}$ kg (1); Final statement (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 65°	The two acute angles add up to 90°.
2	**(b)** $\frac{3}{14}$	Multiply numerators and multiply denominators, then simplify.
3	**(a)** 25°	Complementary angles add up to 90°.
4	**(b)** odd	Use the even/odd rules.
5	**(a)** No	Two sides together must be longer than the third.
6	**(a)** 12	'Of' means multiply.
7	**(b)** 65°	Supplementary angles add up to 180°.
8	**(c)** 16	Add all the digits.
9	**(b)** 70°	The three angles add up to 180°.
10	**(c)** $\frac{2}{5}$	Turn the fraction upside down.
11	**(c)** 75°	Vertically opposite angles are equal.
12	**(c)** 8021	Write the digits in the opposite order.
13	**(b)** isosceles	Recall the definition.
14	**(a)** $\frac{3}{7}$	Dividing by n means multiplying by $\frac{1}{n}$.
15	**(c)** 80°	A linear pair adds up to 180°.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\angle C = 180^\circ - 105^\circ = 75^\circ$; $\angle ACD = 180^\circ - 75^\circ$ (1); $= 105^\circ$ (equal to $\angle A + \angle B$) (1).	1+1
17	$\frac{3}{5} \times 40$; $= \frac{120}{5}$ (1); Answer 24 kg (1).	1+1
18	Complement $= 90^\circ - 62^\circ = 28^\circ$ (1); Supplement $= 180^\circ - 75^\circ = 105^\circ$ (1).	1+1
19	Apply the rule to each number in turn; Sequence: $5, 16, 8, 4, 2, 1$ (1); It reaches 1 after 5 steps (1).	1+1
20	Two shorter sides: $3 + 5 = 8$; Compare with the longest side 9 (1); No, 8 is not greater than 9 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Total parts $= 9$; one part $= 180^\circ \div 9 = 20^\circ$ (1); Angles 40°, 60°, 80° (1); Check: 180° (1).	1+1+1
22	Improper form: $1\frac{1}{2} = \frac{3}{2}$ (1); $\frac{3}{2} \times \frac{4}{9} = \frac{12}{18}$ (1); Answer $\frac{2}{3}$ (1).	1+1+1
23	$x = 180 - 135 = 45$ (1); The angle 45° is acute (2).	1+2
24	Middle number $= 75 \div 3 = 25$ (1); The numbers are 24, 25 and 26 (1); Check: $24 + 25 + 26 = 75$ (1).	1+1+1
	OR: Both base angles are 70°, sum $= 140^\circ$ (1); Apex angle $= 180^\circ - 140^\circ$ (1); $= 40^\circ$ (1).	1+1+1
25	Pieces $= n \div \frac{5}{6}$ (1); $10 \times \frac{6}{5} = \frac{60}{5}$ (1); Pieces $= 12$ (1).	1+1+1
	OR: (i) Alternate interior angles are equal: 66° (1); (ii) Interior angles on the same side add up to 180°: 114° (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Vertically opposite angle $= 50^\circ$ (1); The other two angles are 130° each (linear pair) (1); Sum $= 50 + 50 + 130 + 130 = 360$ (1); The four angles around the point add up to 360° (2).	1+1+1+2
	OR: Reverse $= 74$ (1); Sum $= 47 + 74 = 121$ (1); $121 = 11 \times 11$ (1); Divisible by 11; quotient $= 11$ (2).	1+1+1+2
27	$6 + 8 = 14 > 10$, so the triangle exists (1); It is scalene (1); Perimeter $= 6 + 8 + 10$ (1); $= 24$ cm (1); Final statement (1).	1+1+1+1+1
	OR: At first $\frac{5}{8} \times 80 = 50$ L (1); Used $\frac{4}{5} \times 50 = 40$ L (1); Left $= 50 - 40$ (1); $= 10$ L (1); State all three answers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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